Type-Based Unsourced Federated Learning With Client Self-Selection

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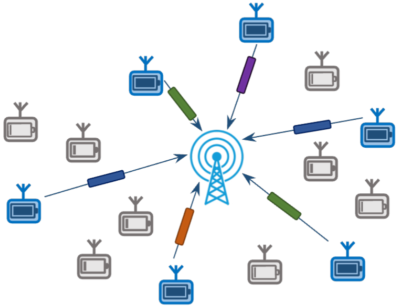
Considered problem setup

- Digital over-the-air federated learning
- Massive number of uncoordinated and sporadically active devices operating over a fading channel
- Data heterogeneity across clients

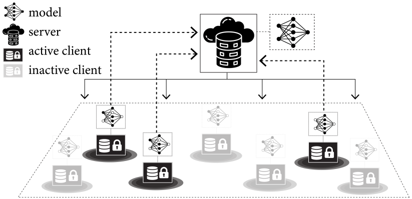


Key research questions:

How can we design a **client selection** + **unsourced transmission scheme** that:

1. preserves client privacy,
2. remains reliable under fading and interference?

Over-the-air federated learning



Federated averaging (FedAvg) [McMahan et al., 2017]:

Clients collaboratively train a shared global model over multiple rounds $t = 1, 2, \ldots, T$:

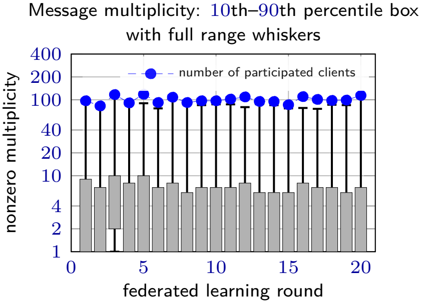
- Server broadcasts the current global model $\boldsymbol{w}^{(t)} \in \mathbb{R}^W$
- Each client performs local training using its own dataset and obtains local model $\boldsymbol{w}_k^{(t)}$
- Local model updates $\boldsymbol{\Delta}_k^{(t)} = \boldsymbol{w}_k^{(t)} - \boldsymbol{w}^{(t)}$ are transmitted over a shared wireless channel
- The server aggregates the received local updates $\{\boldsymbol{\Delta}_k^{(t)}\}$ and obtains new global model $\boldsymbol{w}^{(t+1)}$

Many clients must simultaneously transmit local updates $\{\boldsymbol{\Delta}_k^{(t)}\}$ over a shared wireless medium

Unsourced multiple access with message collisions

Unsourced multiple access (UMA) [Polyanskiy, 2017]:

- All devices employ the **same encoder**
- Receiver produces an **unordered list** of messages
- Standard UMA treats **message collisions as errors**



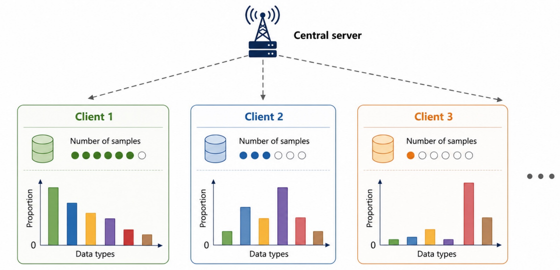
Type-based UMA (TUMA) [Ngo et al., 2024]:

- Multiple users may transmit the **same message**
- In federated learning, the **same quantized local update**
- Interested in both the set of transmitted messages and their **multiplicities**
- **Goal:** estimate empirical message distribution (type)

TUMA solutions over fading channels:

- MD-AirComp scheme [Qiao et al., 2024] requires channel state information at transmitters (CSIT)
- **Our approach:** CSIT-free TUMA over distributed MIMO (D-MIMO) [Okumus et al., 2025]

Data heterogeneity and client selection



- Only a **subset of clients** can participate in each communication round
- **Random client selection performs poorly** under heterogeneous local datasets
- **Existing** methods rely on **server-side client information**, violating privacy
- Power-of-choice (PoC) [Cho et al., 2022]: clients with **larger local training losses** are preferred
- PoC: clients **report local losses** to the server for selection, partially violating privacy

Can we design a **privacy-preserving** client selection strategy (**without server-side client information**)?

Our contributions

We propose:

→ Client self-selection strategy, no server-side client information required

→ TUMA-based model aggregation over D-MIMO fading channels

In this presentation

client self-selection
under perfect communication

TUMA in a D-MIMO network
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Our proposed client self-selection strategy

Key ideas:

- Inspired by PoC [Cho et al., 2022]: clients with higher local loss participate with higher probability
- Unlike PoC, self-selection is performed locally at clients
- The server adjusts a global threshold to control the number of participating clients

Self-selection procedure (per round t): target number of selected clients K_{tar}

1. **Candidate set:** Active clients enter the candidate set randomly
2. **Self-selection:** Candidate client k computes its loss $\ell_k^{(t)}$ and participates with probability

$$\mathrm{sigmoid}\left(a\left(\ell_k^{(t)} - \theta^{(t)}\right)\right)$$

3. **Threshold update:** The server adjusts the threshold as

$$\theta^{(t+1)} \leftarrow \theta^{(t)} + \xi\left(\widehat{L}^{(t)} - K_{\mathrm{tar}}\right),$$

where $\widehat{L}^{(t)}$ is the estimated number of participating clients

Simulation results under perfect communication

image classification over FMNIST dataset, moderate data heterogeneity $\mathsf{Dir}(\alpha = 2)$,
total number of clients $K = 1000$, target number of selected clients $K_{\mathrm{tar}} = 100$

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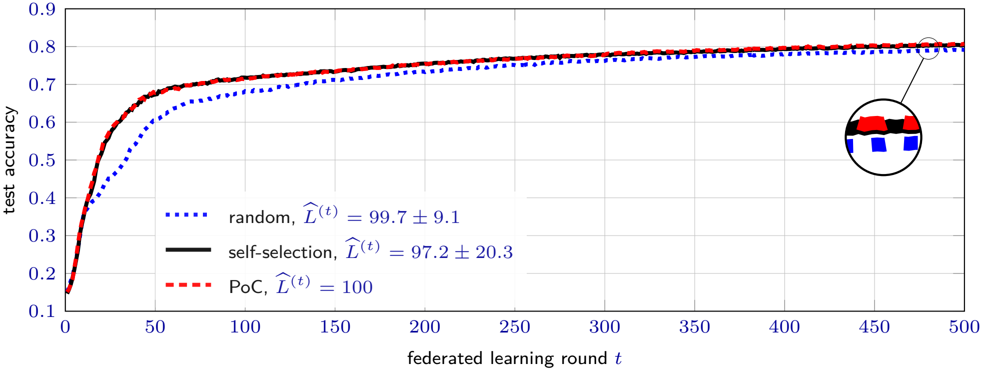

The proposed self-selection works close to the PoC and outperforms random selection

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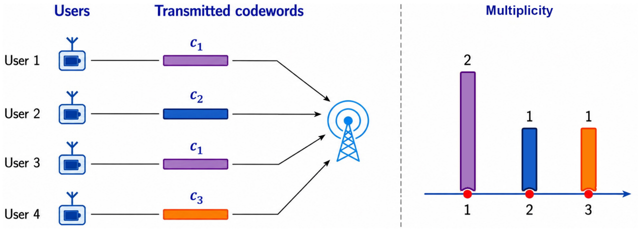
Type-based unsourced federated learning



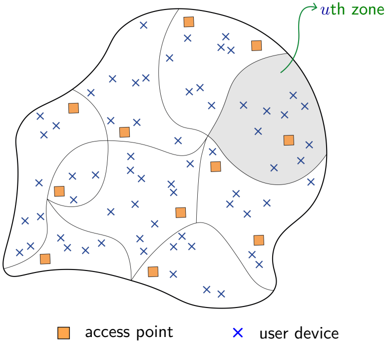
- The model updates are vector-quantized, mapped to codewords and transmitted
- To perform federated learning, the receiver needs to infer which codewords are transmitted and how many times (empirical codeword distribution, i.e., type)

Fading makes it difficult to estimate multiplicities, as we cannot rely on received power

State-of-the-art method: MD-AirComp [Qiao et. al, 2024]

Uses CSIT to perform power control but difficult to implement in practice

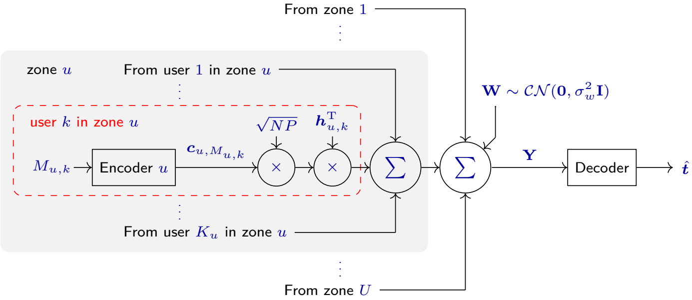
TUMA over a D-MIMO network [Okumus et al., 2025]



Our approach: TUMA over a D-MIMO network

- area partitioned into U zones
- users in a zone have similar large-scale fading coefficients (LSFCs)
- codebook partitioned into U zones
- codebook for zone u: $\mathbf{C}_u \in \mathbb{C}^{N \times 2^J}$
- N: blocklength
- 2^J: number of possible messages
- Access points (APs) with multiple antennas each
- F antennas in total
- APs connected to the server via fronthaul links

System overview of TUMA over D-MIMO [Okumus et al., 2025]



- Model updates are vector-quantized into messages
- Many clients transmit messages (quantized model updates) $\{M_{u,k}\}$ simultaneously
- Fading modeled via channel vector $\boldsymbol{h}_{u,k} \in \mathbb{C}^F$ (from user to all AP antennas)
- Server receives superposition signal $\mathbf{Y}$ and estimates the type vector $\boldsymbol{t}$ from it
- AMP-based algorithm for type estimation

Vector quantization for wireless federated learning

Challenge: High-dimensional model updates

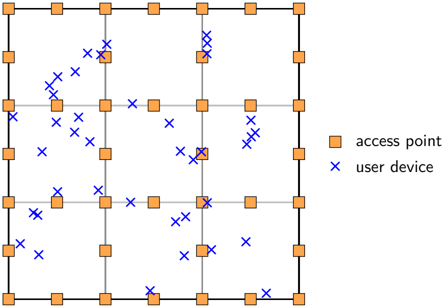
Local model updates $\boldsymbol{\Delta}_k^{(t)} \in \mathbb{R}^W$ cannot be transmitted directly over the wireless channel

Vector quantization with error accumulation

Following [Amiri et al., 2019] and [Qiao et al., 2024]:

- Local model updates are compressed using vector quantization
- Error accumulation mitigates quantization distortion
- Quantization codebook $\mathcal{Q} = \{\boldsymbol{q}_1, \cdots, \boldsymbol{q}_{2^J}\}$ with number of bits J and vector dimension Q
- Quantization indices $\{M_k^{(d,t)}\}_{d=1}^{D}$ are transmitted through TUMA
- Large model dimension requires multiple communication subrounds $D = \lceil W/Q \rceil$

Simulation settings



Example of TUMA with D-MIMO:

- 300 m $\times$ 300 m coverage area
- $U = 9$ zones
- 40 APs with 4 antennas each
- User positions uniformly at random
- LSFC between user at position $\boldsymbol{\rho}$ and AP b at position $\boldsymbol{\nu}_b$ is modeled as

$$\gamma_b(\rho) = \frac{1}{1 + (\|\boldsymbol{\rho} - \boldsymbol{\nu}_b\|/d_0)^{\alpha}},$$

where $\alpha = 3.67$ and $d_0 = 13.57$ m

Simulation results over D-MIMO networks

image classification over FMNIST, self-selection, moderate data heterogeneity $\mathrm{Dir}(\alpha = 2)$, total number of clients $K = 1000$, target number of selected clients $K_{\text{tar}} = 100$, vector dimension $Q = 30$, number of bits $J = 7$ ($2^J > N$: orthogonalization impossible)

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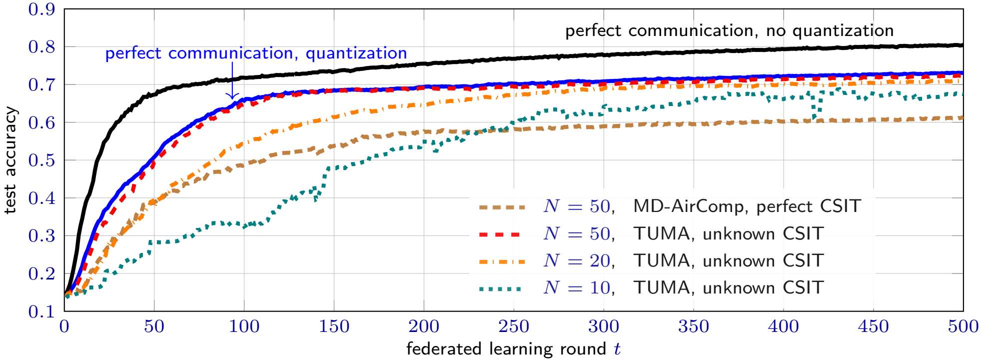
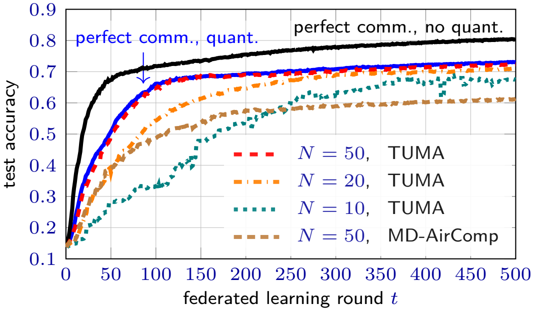


Our proposed method is close to perfect communication with quantization even without CSIT

Conclusion

- We proposed a **privacy-preserving** client self-selection strategy
- Performance **close to PoC** under **data heterogeneity**
- We applied **TUMA over D-MIMO** for wireless federated learning
- Improved performance over **MD-AirComp** without requiring **CSIT**



Code and examples available at:
`github.com/okumuskaan/ufl_tuma`